



Pd supported catalyst for gas-phase 1,2-dichloroethane abatement: Efficiency and high selectivity towards oxygenated products



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ABSTRACT

Hydroxyapatite (HAP) support has been synthesised and impregnated with different amounts of Pd. The prepared Pd/HAP catalysts have been thoroughly characterised by BET, XRD, TEM, UV–visible–NIR, FTIR, XPS, CO₂-TPD and NH₃-TPD techniques and tested in the DCE oxidation reaction.

The characterisation of the catalysts has revealed a clear evolution of the Pd species structure, by increasing the Pd content, from highly dispersed tetrahedral coordinated Pd²⁺ to larger Pd²⁺ species particles adopting square planar geometry. The latter seems to be partially encapsulated by the HAP support.

In DCE oxidation reaction the Pd/HAP system has shown an activity comparable to that of conventional system (Pd/alumina). However, the former has proved high selectivity towards the production of oxygenated products (CO_x (CO₂ and CO)). Notably, in contrast to Pd/alumina, the DCE oxidation does not yield vinyl chloride intermediate product over Pd/HAP. This catalytic behaviour has been related to the moderate acidity of HAP compared to that of alumina together with its interaction with the Pd active phase. This report, then, considers that Pd(x)/HAP could be presented as a good alternative to those reported in the available literature.

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Introduction

Air pollution caused by the release of volatile organic compounds (VOCs) in different industrial activities is currently considered one of the major environmental problems [1–3]. In this sense, chlorine-containing organic compounds (Cl-VOCs), which are produced in considerable amounts nowadays, require special attention; especially considering their high volatility, toxicity and stability. These chlorinated VOCs, which include polychlorinated methanes, ethanes and ethylenes, are essentially used as cleaning and extraction agents, additives for adhesives and solvents for chemical reactions in a number of industrial processes.

For a long time, thermal incineration has been employed as an effective remediation technology for the removal of Cl-VOCs. There are, however, a number of drawbacks that strongly restrict the application of traditional incinerators as they require high

operation temperatures and release undesirable by-products [1–3]. For this reason, in the last few decades, research has focused on the development of catalytic processes that could allow operating at low temperatures in order to decrease the processing cost and NO_x emissions. Recently, much effort has been devoted to the development of catalytic systems for hydrodechlorination processes to transform Cl-VOCs to unsaturated hydrocarbons with high added value [4–9]. However, the economic viability of this strategy, which involves hydrogen consumption and a demanding catalyst activation step, lacks evidence to be considered as an alternative. By contrast, catalytic oxidation of Cl-VOCs is still considered as a method of choice because it allows the oxidation to proceed at low temperatures and results in a diminution of NO_x emissions [1–3]. Moreover the use of efficient catalysts can increase the selectivity towards deep oxidation products (CO₂ and HCl) at the expense of undesirable products, such as CO, vinyl chloride (VC) and Cl₂.

Generally, catalysts for Cl-VOCs oxidation consist of transition metal oxides (Co, V, Cr, Mn) and noble metal (Pt, Pd, Ru and Rh) supported on alumina, zeolite and ceria [1–3,10–16].

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